

Multiple Choice (10 marks)

1. Which types of functions grow the slowest?
 - (a) $O(\log e^N)$
 - (b) $O(\log N)$
 - (c) $O(\log \log N)$
 - (d) $O(N)$
2. In Python 3, which of the following function sets the integer starting value used in generating random numbers?
 - (a) `choice(seq)`
 - (b) `randrange ([start,] stop [,step])`
 - (c) `random()`
 - (d) `seed([x])`
3. Of the following sorting algorithms, which has a running time that is LEAST dependent on the initial ordering of the input?
 - (a) Insertion sort
 - (b) Quicksort
 - (c) Merge sort
 - (d) Selection sort
4. Which types of functions grow the slowest?
 - (a) $O(N^{1/2})$
 - (b) $O(N^{1/4})$
 - (c) $O(N^{1/N})$
 - (d) $O(N)$
5. Let $x = 1$. What is $x \ll 3$ in Python 3?
 - (a) 1
 - (b) 3
 - (c) 8
 - (d) 16

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, accessing the first element of the tuple ('abcd', 786 , 2.23, 'john', 70.2) results in an error.
7. True or False: The one-time pad is considered a perfectly secure encryption scheme when used correctly with truly random keys.
8. True or False: Adding the two's-complement numbers 10000001 and 10101010 will result in an overflow condition in 8-bit representation.
9. True or False: In Python 3, the output of `print(list[1:3])` is `[786, 2.23]` when `list = ['abcd', 786, 2.23, 'john', 70.2]`.
10. True or False: In a complete K-ary tree of depth N, the ratio of nonterminal nodes to total nodes approximates $(K-1)/K$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to convert a date of yyyy-mm-dd format to dd-mm-yyyy format.
12. Write a function to find out the number of ways of painting the fence such that at most 2 adjacent posts have the same color for the given fence with n posts and k colors.

End of Paper